The lesson to me seems clear: There may be no way to send humans to Mars in the comparatively near future—despite the fact that it is entirely within our technological capability. Governments do not spend these vast sums just for science, or merely to explore. They need another purpose, and it must make real political sense.

It may be impossible to go just yet, but when it is possible, the mission, I think, must be international from the start, with costs and responsibilities equitably shared and the expertise of many nations tapped; the price must be reasonable; the time from approval to launch must fit within practical political timescales; and the space agencies concerned must demonstrate their ability to muster pioneering exploratory missions with human crews safely, on time, and on budget. If it were possible to imagine such a mission for less than $100 billion, and for a time from approval to launch less than 15 years, maybe it would be feasible. (In terms of cost, this would represent only a fraction of the annual civilian space budgets of the present spacefaring nations.) With aerobraking and manufacturing fuel and oxygen for the return trip out of Martian air, it's now beginning to look as if such a budget and such a timescale might actually be realistic.

The cheaper and quicker the mission is, necessarily the more risk we must be willing to take with the lives of the astronauts and cosmonauts aboard. But as is illustrated, among countless examples, by the samurai of medieval Japan, there are always competent volunteers for highly dangerous missions in what is perceived as a great cause. No budget, no timeline can be really reliable when we attempt to do something on so grand a scale, something that has never been done before. The more leeway we ask, the greater is the cost and the longer it takes to get there. Finding the right compromise between political feasibility and mission success may be tricky.

IT'S NOT ENOUGH to go to Mars because some of us have dreamt of doing so since childhood, or because it seems to us the obvious long-term